Arista Access Node BTM-ACC-19 QoS Congestion Event Resolution SOP

Node: BTM-ACC-19 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_9298

Description

This Standard Operating Procedure (SOP) outlines the steps to mitigate the QoS congestion event on the Arista Access Node BTM-ACC-19, which is experiencing microbursts, interface tail drops, and QoS queue starvation, indicated by the ALARM_9298 alarm code with a severity of Warning. This failure domain occurs on this specific role and vendor equipment due to the inherent limitations of the egress queue management mechanism, which is based on the Deficit Weighted Round-Robin (DWRR) and Weighted Random Early Detection (WRED) algorithms. The DWRR algorithm is prone to queue starvation in scenarios with high traffic variability, leading to microbursts and interface tail drops. Furthermore, the WRED algorithm may not effectively mitigate congestion in cases where the bursty traffic exceeds the configured threshold. As a result, the QoS queue may become overwhelmed, triggering the ALARM_9298 alarm code. This SOP aims to provide a step-by-step guide for diagnosing and resolving the QoS congestion event on the Arista Access Node BTM-ACC-19.

Pre-requisite Checks

- Verify that the remote hands are dispatched to BTM Layout Aggregation to provide physical access to the node.

- Check the status of the redundant link to ensure it is functioning correctly and not contributing to the congestion.

- Gather relevant network traffic statistics and logs to analyze the traffic patterns and identify potential causes of congestion.

Potential Root Causes

Cause	Probability
DWRR algorithm-induced queue starvation	High
WRED algorithm failure to mitigate congestion	Medium
High traffic variability causing microbursts	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface status, Expected Output: Interface status: up/down, bandwidth: 1Gbps/10Gbps, Description: This command displays the interface status and bandwidth to verify that the interface is functioning correctly.

Command: show queue statistics, Expected Output: Queue statistics: dropped packets, queue depth: 1000/5000, Description: This command
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```
displays the queue statistics to identify the number of dropped
packets and queue depth, which can indicate congestion.

Command: show qos policy, Expected Output: QoS policy: DWRR algorithm,
WRED algorithm, queue depth: 5000, Description: This command displays
the QoS policy to verify that the DWRR and WRED algorithms are
configured correctly and to identify the queue depth.
```

Resolution Steps

Step 1: Step: Adjust DWRR algorithm parameters, Command: config qos policy dwrr-params adjusted , Description: Adjust the DWRR algorithm parameters, such as the queue depth and bandwidth, to mitigate queue starvation and improve congestion handling.

Step 2: Step: Configure WRED algorithm, Command: config qos policy wred-params enabled, Description: Configure the WRED algorithm to effectively mitigate congestion and reduce the likelihood of queue starvation.

Step 3: Step: Adjust queue depth, Command: config qos policy queue-depth-adjusted, Description: Adjust the queue depth to reduce the likelihood of queue starvation and improve congestion handling.

Post-Check Verification

```
Command: show interface status, Expected Output: Interface status:
up/up, bandwidth: 1Gbps/10Gbps, Description: Verify that the interface
status has improved after the resolution steps.

Command: show queue statistics, Expected Output: Queue statistics:
dropped packets: 0, queue depth: 1000, Description: Verify that the
queue statistics have improved after the resolution steps.

Command: show qos policy, Expected Output: QoS policy: DWRR algorithm,
WRED algorithm, queue depth: 1000, Description: Verify that the QoS
policy has improved after the resolution steps.
```

Roll-back Procedures

- Step: Restore original DWRR algorithm parameters, Command: config qos policy dwrr-params restore, Description: Restore the original DWRR algorithm parameters to their default values if the resolution steps fail.

- Step: Disable WRED algorithm, Command: config qos policy wred-params disabled, Description: Disable the WRED algorithm if the resolution steps fail.

- Step: Restore original queue depth, Command: config qos policy queue-depth-original, Description: Restore the original queue depth to its default value if the resolution steps fail.